Assignment Day 1

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Day 1. Tools for troubleshooting theory :

Task 1. list three real-life scenarios where a multimeters would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

→ Dead phone not charging :-
measure voltage across USB port connector to reveal if charging port is receiving power

* computer won't turn on :
check DC voltage output from the power supply unit (PSU) cables to confirm it is delivering the right power to the motherboard.

* Short circuit on a motherboard :-
Tests for continuity across tiny board components (like capacitors) to pinpoint where electrical current is leaking to ground.

2. watch a short youtube video demo of a post card in action, then brief a summary (3-4 lines) describing what the postcard displays and what problem it helped identify in the video.

→ in the video post PCI card Diagnostic RAM is not correctly seated.

* what it displays :-
A two digit LED screen that shows hexadecimal diagnostic codes along with red status indicator lights during the boot check.

* Error code shown :-
The card stops cycling through start up checks and halts directly on code d4.

* problem identified :-
code d4 points to a memory initialization failure, revealing Ram stick was not proper lock in its slot on motherboard.

3. Download and install any free software diagnostic tool run it on your laptop / Desktop and take a screenshot of main dashboard showing system health details.

File Edit Function Theme Disk Help Language

Caution
40 °C
C:

IC25N030ATCS04-0 : 30.0 GB

Health Status

Caution

Temperature

40 °C

Firmware	CA3OA71A
Serial Number	***************
Interface	Parallel ATA
Transfer Mode	UDMA/100 | UDMA/100
Drive Letter	C:
Standard	ATA/ATAPI-5 | ATA/ATAPI-5 T13 1321D version 3
Features	S.M.A.R.T., APM

Buffer Size	1768 KB
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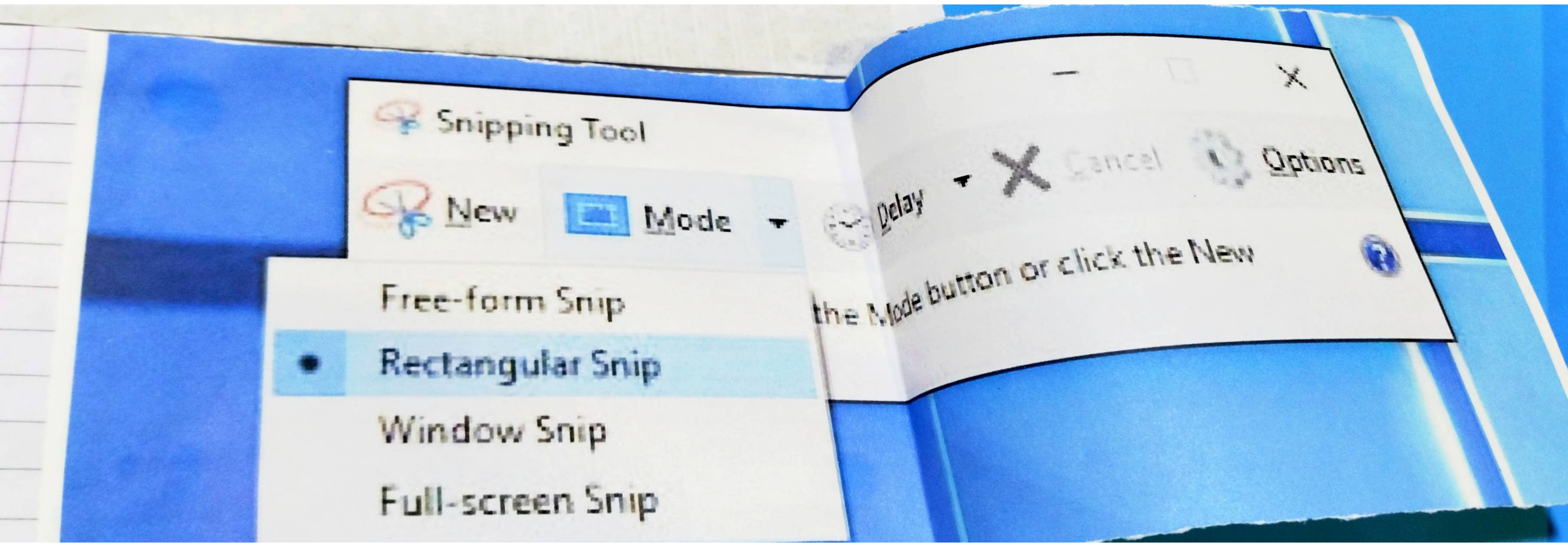
Power On Count	1319 count
Power On Hours	5369 hours

ID	Attribute Name	Current	Worst	Threshold	Raw Values
01	Read Error Rate	100	100	62	000000000000
02	Throughput Performance	100	100	40	000000000000
03	Spin-Up Time	121	121	33	001200000001
04	Start/Stop Count	100	100	0	0000000005B3
05	Reallocated Sectors Count	100	100	5	000000000000
07	Seek Error Rate	100	100	67	000000000000
08	Seek Time Performance	100	100	40	000000000000
09	Power-On Hours	88	88	0	0000000014F9
0A	Spin Retry Count	100	100	60	000000000000
0C	Power Cycle Count	100	100	0	000000000527
BF	G-Sense Error Rate	100	100	0	000000010000
C0	Power-off Retract Count	100	100	0	00000000002F
C1	Load/Unload Cycle Count	90	90	0	000000019318
C2	Temperature	137	137	0	003D00080028
C4	Reallocation Event Count	100	100	0	00000000000F
C5	Current Pending Sector Count	100	100	0	000000000008
C6	Uncorrectable Sector Count	100	100	0	000000000000
C7	UltraDMA CRC Error Count	200	200	0	000000000000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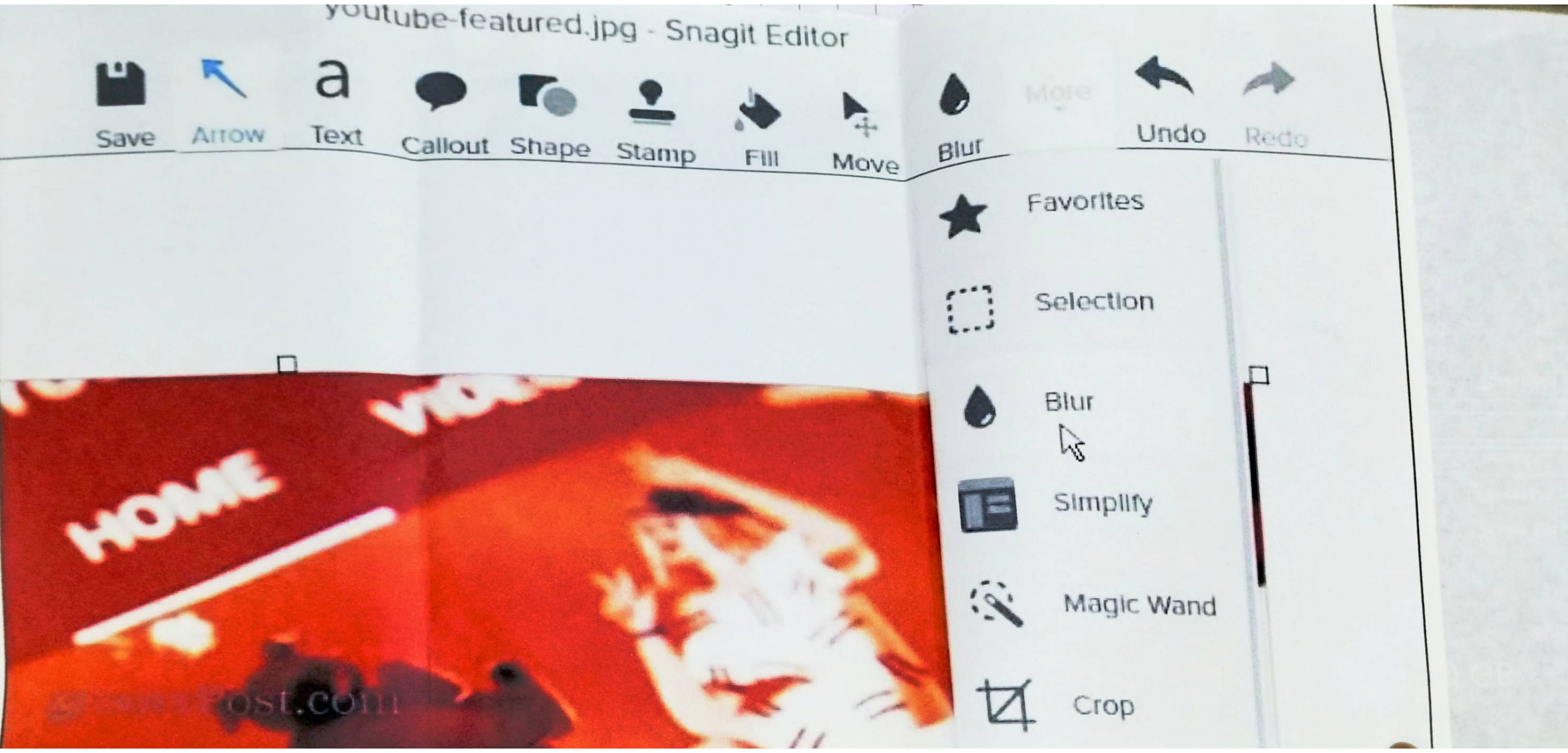
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4. Compare the use of a multimeter, POST card and software diagnostics by creating a table with three columns what it checks, Example problem and fill in at least one example for each tool.

→

Tool	What it checks	Example problem
multimeter	Physical electricity, voltage level and circuit countinuity in power components & wires	Dead power supply output blown fuse or Short circuit.
POST card	motherboard initialization steps during early boot before the screen turn on.	Bad RAM or broken motherboard trace preventing system startup.
Software Diagnostics	component health stability & system performance while the OS or test environment is running	Hard drive with bad sectors or CPU overheating under heavy load.